PAPER_1079: Galaxy Cluster Cooling-Flow Buoyancy Suppression

Star Magic UQFF Framework — Session 225

Author: Daniel Murphy **Date:** 2026 **Module:** `galaxy_{cluster\_cooling\_flow}.py`

Abstract

We present a parameterized calculator for galaxy-cluster cooling-flow regulation via $F_{U,Bi,i}$ jet-mediated AGN feedback. The model balances intra-cluster medium (ICM) radiative cooling against buoyancy-driven jet heating to predict the cooling-flow suppression factor $S(\Gamma)$ as a function of SCm phonon linewidth.

1. ICM Radiative Cooling

The cooling luminosity of the ICM is dominated by thermal bremsstrahlung:

$$L_{\text{cool}} = n_e^2 \cdot \Lambda(T) \cdot V_{\text{cool}}$$

where the cooling function for metallicity $Z = 0.3$ solar:

$$\Lambda_{\text{ff}}(T) = 1.42 \times 10^{-40} \cdot \sqrt{T} \cdot g_{\text{ff}} \cdot (1 + 1.8Z) \quad [\text{W m}^3]$$

The classical mass deposition rate without AGN feedback:

$$\dot{M}_{\text{cool}} = \frac{L_{\text{cool}} \cdot \mu m_p}{(5/2) k_B T}$$

2. $F_{U,Bi,i}$ Jet Heating

The AGN jet mechanical power is linked to the SMBH through the 26-layer buoyancy force:

$$F_{U,Bi,i} = \sum_{i=1}^{26} [Ug_i - Ub_i] + U_m + U_A + F_n \cdot S_{26}^{(3)} \cdot \Phi(\Gamma) \cdot E_{\text{net}}$$

The jet heating power:

$$Q_{\text{heat}} = |F_{U,Bi,i}| \cdot \eta_{\text{jet}} \cdot M_{\text{ICM}} \cdot v_{\text{jet}}$$

3. Cooling-Flow Suppression Factor

$$S(\Gamma) = \min\left(\frac{Q_{\text{heat}}(\Gamma)}{L_{\text{cool}}}, 1\right)$$

The suppressed mass deposition rate:

$$\dot{M}_{\text{suppressed}} = \dot{M}_{\text{classical}} \times (1 - S)$$

When $S \to 1$: complete cooling-flow quenching (AGN dominates). When $S \to 0$: no suppression (classical cooling catastrophe).

4. Cluster Applications

Perseus (NGC 1275)

- $T_{\text{ICM}} = 4.0$ keV, $n_e = 3 \times 10^4$ m^{-3}, $r_{\text{cool}} = 100$ kpc
- $M_{\text{BH}} = 3.4 \times 10^8$ M$_\odot$
- $L_{\text{cool}} = 1.98 \times 10^{38}$ W
- $t_{\text{cool}} = 0.57$ Gyr
- $S = 1.0$ (fully suppressed)

Abell 2256

- $T_{\text{ICM}} = 6.5$ keV, merging cluster
- $M_{\text{BH}} = 10^9$ M$_\odot$

Virgo (M87)

- $T_{\text{ICM}} = 2.5$ keV, relativistic jet ($v_{\text{jet}} = 0.99c$)
- $M_{\text{BH}} = 6.5 \times 10^9$ M$_\odot$

5. Gamma Dependence

The suppression factor varies with phonon linewidth through $\Phi(\Gamma)$ in the $F_{U,Bi,i}$ force. A sweep over $\Gamma \in [0.01, 0.50]$ THz shows peak suppression near $\Gamma_0 = 0.10$ THz, consistent with the SCm resonance framework.

6. Validation

- All 3 clusters produce valid suppression factors $S \in [0, 1]$
- Cooling times physically realistic (0.01 - 1000 Gyr)
- 10/10 self-tests pass

References

1. Perseus cooling flow: NGC 1275, $M_{\text{cool}} \sim 13 \times 10^9$ M$_\odot$ H$_2$
2. Sutherland & Dopita (1993): Cooling functions
3. $F_{U,Bi,i}$ framework: `MAIN_{1\_CoAnQi}.cpp` SOURCE4
4. ICM physics: `gen_{muge\_ngc1275}.py`

Session 225: Late-Corpus Physics Integration (PAPER_1000-1081)

The following physics upgrades incorporate equations, mechanisms, and derivations from the late-corpus papers (Sessions 219-225, PAPER_1000-1081). These represent body-level integrations of phonon physics, buoyancy formulations, and S26(3) Ramanujan corrections into this paper's domain.

Session 225 Phonon-Physics Upgrade: ICM Buoyancy Force Profile

Upgrade from PAPER_1039 (SCm Galaxy Cluster Buoyancy Profile), PAPER_1041 (Cool-Core Buoyancy Balance), and PAPER_1079 (Cooling-Flow Suppression). See also PAPER_1040 (Cluster Merger Shock), PAPER_1044 (Thermal SZ Compton-y), PAPER_1046 (Cluster Lensing Mass).

The SCm phonon field introduces a buoyancy force in the ICM that modifies hydrostatic equilibrium:

$$F_{\text{buoy}}(r) = \rho(r) \cdot V \cdot g(r) \cdot \beta_i \cdot S_{26} \cdot \Phi$$

where the ICM density follows the beta-model:

$$\rho(r) = \rho_0 \left(1 + \left(\frac{r}{r_c}\right)^2\right)^{-3\beta/2}$$

Hydrostatic mass bias reduction (PAPER_1039):

$$b_{\text{UQFF}} = 1 - \frac{M_{\text{HSE}}}{M_{\text{true}}} = 0.17 \qquad (\text{vs standard } b = 0.20)$$

The buoyancy pressure contributes $P_{\text{buoy}}/P_{\text{thermal}} \approx 3$–$4\%$ at cluster cores, partially resolving the Planck SZ-CMB mass tension.

Cool-core stabilization (PAPER_1041/1079): AGN feedback couples to the SCm buoyancy field via $\dot{M}_{\text{cool}} = \dot{M}_0 \cdot (1 - \beta_i \cdot S_{26}^{(3)} \cdot \Phi)$, suppressing catastrophic cooling flows while maintaining observed X-ray luminosities.

Phonon frequency coupling: $\omega_{\text{SCm}} = 2\pi \times 1.25$ THz sets the temporal scale for buoyancy oscillations; the ratio $\omega_{\text{SCm}}/\omega_{\text{sound}}$ governs the phonon transmission efficiency across the ICM.

Session 225 Phonon-Physics Upgrade: S26(3) Ramanujan Summation

Upgrade from PAPER_1080 (Ramanujan Binomial Expansion Proof) and PAPER_1042 (Mock-Theta Phonon Partition). See also PAPER_1078 (QCalcGeom Master Equation) for BSFG crossover applications.

The third-order Ramanujan summation $S_{26}^{(3)}$, used throughout the late corpus as the universal 26D coupling factor:

$$S_{26}^{(3)} = \sum_{n=0}^{\infty} \frac{(1/4)_n\,(1/2)_n\,(3/4)_n}{(n!)^3} \cdot \prod_{i=1}^{26} \left[1 + [\text{SSq}] \cdot e^{-\kappa\, i\, n/26}\right]$$

where $(a)_n = a(a+1) \cdot s(a+n-1)$ is the Pochhammer symbol.

Binomial expansion (PAPER_1080): The convergence proof shows:

$$R_n^{(26,3)} = \binom{4n}{n} \cdot \frac{W_{26}(n)}{(4^{4n})} \qquad \text{with} \quad W_{26}(n) = \prod_{i=1}^{26} \left[1 + [\text{SSq}] \cdot e^{-\kappa\, i\, n/26}\right]$$

This sum converges absolutely for $|[\text{SSq}]| < 1$ (satisfied by $[\text{SSq}] = 0.57$) and reduces to the classical Ramanujan $1/\pi$ series when $[\text{SSq}] \to 0$.

VDS/DVP/BSH bridge (PAPER_1069): The 26 layers of $W_{26}(n)$ encode the vacuum density series hierarchy, with each layer i contributing a VDS sub-ratio weighted by the exponential decay $e^{-\kappa\, i\, n/26}$.

Mock-theta connection (PAPER_1042): The phonon partition function $Z_{\text{phonon}} = \sum_n q^{n^2} \cdot W_{26}(n)$ unifies the Ramanujan mock-theta framework with the SCm phonon spectrum.

Appendix: Session 225 Cross-References (PAPER_1000–1081)

Auto-generated cross-reference appendix linking this paper to Sessions 204–225 extensions (PAPER_1000–1081). Added by `update_{corpus\_crossrefs}.py` *(Session 225, April 2026).*

Paper	Title
PAPER_1022	GW Phonon Strain SCm Modulation of h(t)
PAPER_1039	SCm Galaxy Cluster Buoyancy Profile ICM Beta-Model
PAPER_1041	SCm Cool-Core Buoyancy Balance AGN Feedback
PAPER_1033	Galactic Bar Resonance SCm Pattern Speed

4 cross-reference(s) identified.